POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5043D | Semester V | Theory | 4 Credits

Advanced Microprocessors

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5043D as Advanced Microprocessors in Semester V for Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Kharagpur’s Microprocessors course and polytechnic advanced-microprocessor curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

> **Source treatment:** Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. System designs, assembly code, hardware interfacing and kernel-level programming must be based on approved engineering plans, certified hardware data, current architectural standards and competent professional review.

> **Verified source note:** Verified sources support x86 evolution, memory management, Pentium features, and ARM basics. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Advanced Microprocessors studies the architecture, programming and interfacing of high-performance computing units beyond basic 8-bit systems. It develops the ability to analyze x86 architecture evolution (8086 to Pentium), evaluate memory management and protected mode operations, understand multicore and RISC (ARM) principles, and coordinate microprocessor-based solutions while respecting the technical and computational boundaries of electronic and computer engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Digital electronics, computer organization and basic 8-bit microprocessors (e.g., 8085).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the architecture and evolution of the Intel x86 microprocessor family.
2. Analyze the operation of advanced features like paging, multitasking and protected mode.
3. Evaluate the superscalar architecture and features of Pentium and modern multicore processors.
4. Explain the systematic principles of RISC architecture and the fundamentals of ARM processors.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the architecture and memory addressing of x86 microprocessors.	Understand
CO2	Analyze advanced microprocessor features like protected mode and multitasking.	Apply
CO3	Analyze the superscalar and multicore features of modern Intel processors.	Apply
CO4	Explain the architecture and instruction set principles of ARM processors.	Understand

Official CO-PO articulation matrix							
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	Title	Official major topics	Hours	CO	Handbook section
1	x86 Architecture and Evolution	Review of 8086 architecture; Memory segmentation; 80286 features; Introduction to 80386: Register organization, Data types, Addressing modes.	Approx. 14 hours	CO1	Module 1
2	Protected Mode and Memory Management	Protected mode operation: Segmentation, GDT, LDT, IDT; Paging mechanism: Page tables, TLB; Multitasking and Protection levels (Privilege rings); Interrupt handling in protected mode.	Approx. 16 hours	CO2	Module 2
3	Pentium and Multicore Processors	Pentium architecture: Superscalar, Branch prediction, Pipeline; Pentium Pro and MMX features; Introduction to Multicore: Hyper-threading, Multi-core vs Multi-processor; Cache coherence.	Approx. 16 hours	CO3	Module 3
4	RISC and ARM Architecture	CISC vs RISC principles; ARM architecture overview: Register set, CPSR, Pipeline; ARM instruction set basics: Data processing, Branch, Load/Store instructions; Thumb mode.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

x86 Architecture and Evolution

Review of 8086 architecture; Memory segmentation; 80286 features; Introduction to 80386: Register organization, Data types, Addressing modes.

CO1 · Approx. 14 hours

Protected Mode and Memory Management

Protected mode operation: Segmentation, GDT, LDT, IDT; Paging mechanism: Page tables, TLB; Multitasking and Protection levels (Privilege rings); Interrupt handling in protected mode.

CO2 · Approx. 16 hours

Pentium and Multicore Processors

Pentium architecture: Superscalar, Branch prediction, Pipeline; Pentium Pro and MMX features; Introduction to Multicore: Hyper-threading, Multi-core vs Multi-processor; Cache coherence.

CO3 · Approx. 16 hours

RISC and ARM Architecture

CISC vs RISC principles; ARM architecture overview: Register set, CPSR, Pipeline; ARM instruction set basics: Data processing, Branch, Load/Store instructions; Thumb mode.

CO4 · Approx. 14 hours

MODULE 1

x86 Architecture and Evolution

Review of 8086 architecture; Memory segmentation; 80286 features; Introduction to 80386: Register organization, Data types, Addressing modes.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

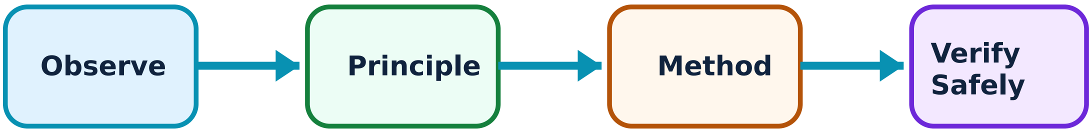
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for x86 Architecture and Evolution: identify the system, state its principle, follow the correct method and verify the result safely.

1. x86 Family Overview−

The x86 family evolved from the 16-bit 8086 to 32-bit and 64-bit architectures. The 8086 introduced 'Memory Segmentation' to address 1MB using 16-bit registers. The 80286 added 'Protected Mode' for multitasking, while the 80386 became the first true 32-bit processor in the line.

Study and application method

Describe 'Memory Segmentation' in 8086; list two improvements of the '80286' over the 8086.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe 'Memory Segmentation' in 8086; list two improvements of the '80286' over the 8086.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Memory segmentation can lead to fragmentation. Do not design memory-intensive applications without authorised consideration of segment limits and base-address calculation.

2. 80386 Microprocessor−

The 80386 introduced 32-bit registers (EAX, EBX, etc.) and a 4GB address space. It supports three operating modes: Real Mode (8086 compatible), Protected Mode (full 32-bit features), and Virtual 8086 Mode (running multiple 8086 tasks).

Study and application method

Sketch the 'Register Organization' of the 80386; explain the purpose of 'Virtual 8086 Mode'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'Register Organization' of the 80386; explain the purpose of 'Virtual 8086 Mode'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Mode switching in x86 is complex. All system-level code must follow authorised sequence for GDT (Global Descriptor Table) initialization before entering protected mode.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Protected Mode and Memory Management

Protected mode operation: Segmentation, GDT, LDT, IDT; Paging mechanism: Page tables, TLB; Multitasking and Protection levels (Privilege rings); Interrupt handling in protected mode.

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

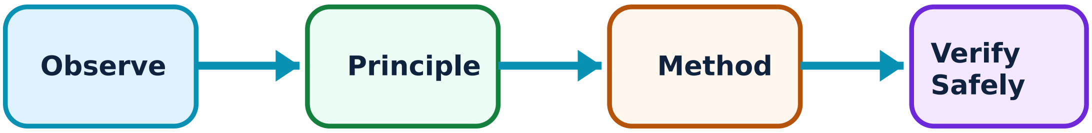
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Protected Mode and Memory Management: identify the system, state its principle, follow the correct method and verify the result safely.

1. Segmentation and Paging−

Protected mode uses descriptors in tables (GDT/LDT) to manage memory and provide protection. 'Paging' divides physical memory into fixed-size pages, allowing for virtual memory management. The 'TLB' (Translation Lookaside Buffer) speeds up address translation.

Study and application method

Explain the role of the 'Global Descriptor Table' (GDT); describe the 'Paging' mechanism conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the role of the 'Global Descriptor Table' (GDT); describe the 'Paging' mechanism conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect page table configuration can cause 'Page Faults' or system crashes. Do not modify kernel-level memory maps without authorised verification of page attributes and permissions.

2. Protection and Multitasking−

The x86 architecture uses four 'Privilege Rings' (0 to 3) to protect the kernel from user applications. Multitasking is supported via 'Task State Segments' (TSS) that store the processor state during context switching.

Study and application method

Describe the 'Privilege Level' (Ring) structure in x86; explain how 'Context Switching' is performed conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the 'Privilege Level' (Ring) structure in x86; explain how 'Context Switching' is performed conceptually.

> **Answer habit:** show a labelled step, the relevant unit/condition and a short interpretation.

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> **Common mistake / precaution:** Privilege escalation is a major security risk. All OS-level code must follow authorised 'Gate' mechanisms (Call Gates, Interrupt Gates) for crossing privilege levels.

> **Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Pentium and Multicore Processors

Pentium architecture: Superscalar, Branch prediction, Pipeline; Pentium Pro and MMX features; Introduction to Multicore: Hyper-threading, Multi-core vs Multi-processor; Cache coherence.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

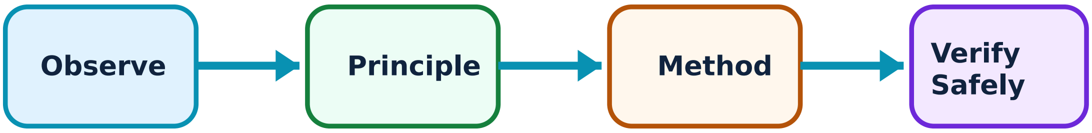
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Pentium and Multicore Processors: identify the system, state its principle, follow the correct method and verify the result safely.

1. Pentium Superscalar Architecture−

The Pentium processor introduced 'Superscalar' execution, allowing two instructions to be executed per clock cycle using two pipelines (U and V). It also features 'Branch Prediction' to minimize pipeline stalls and 'MMX' (Multimedia Extensions) for accelerated graphics and audio.

Study and application method

Define 'Superscalar' architecture; explain the purpose of 'Branch Prediction' in high-speed processors.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Superscalar' architecture; explain the purpose of 'Branch Prediction' in high-speed processors.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Pipeline stalls can significantly reduce performance. Do not optimize assembly code without authorised analysis of instruction dependencies and pipeline hazards.

2. Multicore and Cache Concepts−

Modern processors use multiple cores on a single chip to increase performance while managing power. 'Hyper-threading' allows a single core to act as two logical processors. 'Cache Coherence' protocols (like MESI) ensure that all cores see the same data in their local caches.

Study and application method

Compare 'Multi-core' and 'Multi-processor' systems in a conceptual table; explain the concept of 'Hyper-threading'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'Multi-core' and 'Multi-processor' systems in a conceptual table; explain the concept of 'Hyper-threading'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Cache misses and coherence traffic can bottleneck multicore systems. All parallel software must follow authorised data-locality and synchronization strategies.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

RISC and ARM Architecture

CISC vs RISC principles; ARM architecture overview: Register set, CPSR, Pipeline; ARM instruction set basics: Data processing, Branch, Load/Store instructions; Thumb mode.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

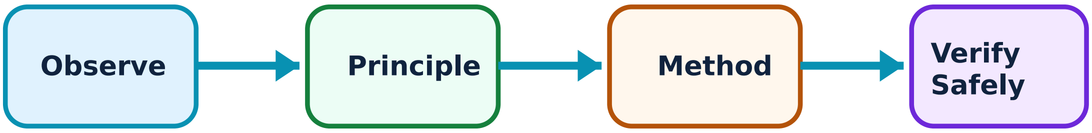
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for RISC and ARM Architecture: identify the system, state its principle, follow the correct method and verify the result safely.

1. RISC Principles and ARM−

RISC (Reduced Instruction Set Computer) focuses on simple, single-cycle instructions to achieve high efficiency. ARM is a popular RISC architecture used in mobile and embedded systems. It features a large register set and a highly efficient pipeline.

Study and application method

Compare 'CISC' and 'RISC' architectures in a conceptual table; list three key features of the 'ARM' architecture.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare 'CISC' and 'RISC' architectures in a conceptual table; list three key features of the 'ARM' architecture.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: RISC architectures rely heavily on compiler optimization. Do not manually optimize RISC assembly without authorised understanding of the underlying pipeline and register-forwarding logic.

2. ARM Programming Model−

ARM uses a 32-bit (or 64-bit) instruction set. The 'CPSR' (Current Program Status Register) tracks processor state. 'Load/Store' architecture means only specific instructions access memory. 'Thumb Mode' provides a 16-bit compressed instruction set for better code density.

Study and application method

Explain the purpose of the 'CPSR' in ARM; describe the advantage of 'Thumb Mode' in embedded applications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the purpose of the 'CPSR' in ARM; describe the advantage of 'Thumb Mode' in embedded applications.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Instruction-set switching (ARM to Thumb) requires careful state management. All firmware must follow authorised branch-and-exchange (BX) protocols for mode transitions.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: x86 Architecture and Evolution

Use the concept flow to revise observation, principle, method and verification.

Module 2: Protected Mode and Memory Management

Use the concept flow to revise observation, principle, method and verification.

Module 3: Pentium and Multicore Processors

Use the concept flow to revise observation, principle, method and verification.

Module 4: RISC and ARM Architecture

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Real Mode and Protected Mode in a conceptual table showing memory and protection differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the 'Paging' address translation process from Virtual to Physical address conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the 'Dual Pipeline' structure of the original Pentium processor showing the U and V pipes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the physical address conceptually for a given segment and offset in 8086 Real Mode.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the processor type and number of cores on your computer or smartphone and note its features.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — x86 Architecture and Evolution

Explain x86 Family Overview and state one correct method, application or precaution. −

The x86 family evolved from the 16-bit 8086 to 32-bit and 64-bit architectures. The 8086 introduced 'Memory Segmentation' to address 1MB using 16-bit registers. The 80286 added 'Protected Mode' for multitasking, while the 80386 became the first true 32-bit processor in the line.

Answer framework: Describe 'Memory Segmentation' in 8086; list two improvements of the '80286' over the 8086.

Important point: Memory segmentation can lead to fragmentation. Do not design memory-intensive applications without authorised consideration of segment limits and base-address calculation.

Explain 80386 Microprocessor and state one correct method, application or precaution. −

The 80386 introduced 32-bit registers (EAX, EBX, etc.) and a 4GB address space. It supports three operating modes: Real Mode (8086 compatible), Protected Mode (full 32-bit features), and Virtual 8086 Mode (running multiple 8086 tasks).

Answer framework: Sketch the 'Register Organization' of the 80386; explain the purpose of 'Virtual 8086 Mode'.

Important point: Mode switching in x86 is complex. All system-level code must follow authorised sequence for GDT (Global Descriptor Table) initialization before entering protected mode.

Module 2 — Protected Mode and Memory Management

Explain Segmentation and Paging and state one correct method, application or precaution. −

Protected mode uses descriptors in tables (GDT/LDT) to manage memory and provide protection. 'Paging' divides physical memory into fixed-size pages, allowing for virtual memory management. The 'TLB' (Translation Lookaside Buffer) speeds up address translation.

Answer framework: Explain the role of the 'Global Descriptor Table' (GDT); describe the 'Paging' mechanism conceptually.

Important point: Incorrect page table configuration can cause 'Page Faults' or system crashes. Do not modify kernel-level memory maps without authorised verification of page attributes and permissions.

Explain Protection and Multitasking and state one correct method, application or precaution.−

The x86 architecture uses four 'Privilege Rings' (0 to 3) to protect the kernel from user applications. Multitasking is supported via 'Task State Segments' (TSS) that store the processor state during context switching.

Answer framework: Describe the 'Privilege Level' (Ring) structure in x86; explain how 'Context Switching' is performed conceptually.

Important point: Privilege escalation is a major security risk. All OS-level code must follow authorised 'Gate' mechanisms (Call Gates, Interrupt Gates) for crossing privilege levels.

Module 3 — Pentium and Multicore Processors

Explain Pentium Superscalar Architecture and state one correct method, application or precaution.−

The Pentium processor introduced 'Superscalar' execution, allowing two instructions to be executed per clock cycle using two pipelines (U and V). It also features 'Branch Prediction' to minimize pipeline stalls and 'MMX' (Multimedia Extensions) for accelerated graphics and audio.

Answer framework: Define 'Superscalar' architecture; explain the purpose of 'Branch Prediction' in high-speed processors.

Important point: Pipeline stalls can significantly reduce performance. Do not optimize assembly code without authorised analysis of instruction dependencies and pipeline hazards.

Explain Multicore and Cache Concepts and state one correct method, application or precaution.−

Modern processors use multiple cores on a single chip to increase performance while managing power. 'Hyper-threading' allows a single core to act as two logical processors. 'Cache Coherence' protocols (like MESI) ensure that all cores see the same data in their local caches.

Answer framework: Compare 'Multi-core' and 'Multi-processor' systems in a conceptual table; explain the concept of 'Hyper-threading'.

Important point: Cache misses and coherence traffic can bottleneck multicore systems. All parallel software must follow authorised data-locality and synchronization strategies.

Module 4 — RISC and ARM Architecture

Explain RISC Principles and ARM and state one correct method, application or precaution.−

RISC (Reduced Instruction Set Computer) focuses on simple, single-cycle instructions to achieve high efficiency. ARM is a popular RISC architecture used in mobile and embedded systems. It features a large register set and a highly efficient pipeline.

Answer framework: Compare 'CISC' and 'RISC' architectures in a conceptual table; list three key features of the 'ARM' architecture.

Important point: RISC architectures rely heavily on compiler optimization. Do not manually optimize RISC assembly without authorised understanding of the underlying pipeline and register-forwarding logic.

Explain ARM Programming Model and state one correct method, application or precaution.−

ARM uses a 32-bit (or 64-bit) instruction set. The 'CPSR' (Current Program Status Register) tracks processor state. 'Load/Store' architecture means only specific instructions access memory. 'Thumb Mode' provides a 16-bit compressed instruction set for better code density.

Answer framework: Explain the purpose of the 'CPSR' in ARM; describe the advantage of 'Thumb Mode' in embedded applications.

Important point: Instruction-set switching (ARM to Thumb) requires careful state management. All firmware must follow authorised branch-and-exchange (BX) protocols for mode transitions.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: x86 Architecture and Evolution.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Protected Mode and Memory Management.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Pentium and Multicore Processors.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.

7. State one key definition from Module 4: RISC and ARM Architecture.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Review of 8086 architecture; Memory segmentation; 80286 features; Introduction to 80386: Register organization, Data types, Addressing modes..
2. Develop a complete answer or practical plan for: Protected mode operation: Segmentation, GDT, LDT, IDT; Paging mechanism: Page tables, TLB; Multitasking and Protection levels (Privilege rings); Interrupt handling in protected mode..
3. Develop a complete answer or practical plan for: Pentium architecture: Superscalar, Branch prediction, Pipeline; Pentium Pro and MMX features; Introduction to Multicore: Hyper-threading, Multi-core vs Multi-processor; Cache coherence..
4. Develop a complete answer or practical plan for: CISC vs RISC principles; ARM architecture overview: Register set, CPSR, Pipeline; ARM instruction set basics: Data processing, Branch, Load/Store instructions; Thumb mode..

LAST-MILE REVISION

Quick Revision

Module 1: x86 Architecture and Evolution

Review of 8086 architecture; Memory segmentation; 80286 features; Introduction to 80386: Register organization, Data types, Addressing modes.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Protected Mode and Memory Management

Protected mode operation: Segmentation, GDT, LDT, IDT; Paging mechanism: Page tables, TLB; Multitasking and Protection levels (Privilege rings); Interrupt handling in protected mode.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Pentium and Multicore Processors

Pentium architecture: Superscalar, Branch prediction, Pipeline; Pentium Pro and MMX features; Introduction to Multicore: Hyper-threading, Multi-core vs Multi-processor; Cache coherence.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: RISC and ARM Architecture

CISC vs RISC principles; ARM architecture overview: Register set, CPSR, Pipeline; ARM instruction set basics: Data processing, Branch, Load/Store instructions; Thumb mode.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
x86 Family Overview	The x86 family evolved from the 16-bit 8086 to 32-bit and 64-bit architectures.
80386 Microprocessor	The 80386 introduced 32-bit registers (EAX, EBX, etc.
Segmentation and Paging	Protected mode uses descriptors in tables (GDT/LDT) to manage memory and provide protection.
Protection and Multitasking	The x86 architecture uses four 'Privilege Rings' (0 to 3) to protect the kernel from user applications.
Pentium Superscalar Architecture	The Pentium processor introduced 'Superscalar' execution, allowing two instructions to be executed per clock cycle using two pipelines (U and V).
Multicore and Cache Concepts	Modern processors use multiple cores on a single chip to increase performance while managing power.
RISC Principles and ARM	RISC (Reduced Instruction Set Computer) focuses on simple, single-cycle instructions to achieve high efficiency.
ARM Programming Model	ARM uses a 32-bit (or 64-bit) instruction set.

LEARNING RESOURCES

References

Recommended advanced microprocessor references

1. Barry B. Brey, The Intel Microprocessors: 8086/8088, 80186/80188, 80286, 80386, 80486, Pentium, Pentium Pro Processor, Pentium II, Pentium III, Pentium 4, and Core2 with 64-bit Extensions.
2. Lyla B. Das, The x86 Microprocessors: 8086 to Pentium, Multicores, Atom and the 8051 Microcontroller.
3. A. K. Ray and K. M. Bhurchandi, Advanced Microprocessors and Peripherals.
4. Steve Furber, ARM System-on-Chip Architecture.

Approved public advanced microprocessor sources

- https://nptel.ac.in/courses/108105102
- https://nptel.ac.in/courses/106108100

These sources provide the verified study basis while official Course 5043D detail is unavailable; they do not replace official chip designs, authorised firmware updates, certified equipment specifications or authorised strategic computing plans.